MIL-G-52903A(ME)
9 May 1985
SUPERSEDING
MIL-G-52903(ME)
15 June 1976

MILITARY SPECIFICATION

GENERATOR SET, DIESEL ENGINE, TRAILER MOUNTED

10 KW, 60 HERTZ, PU-753/M

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a trailer mounted generator set (hereinafter referred to as a power unit) designated PU-753/M. The power unit consists of a Department of Defense design, diesel-engine- driven, 10 kW, 60-Hertz generator set, various auxiliary equipment, brackets, and hardware mounted on a modified Government-furnished trailer chassis.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
QQ-S-781	- Strapping, Steel, and Seals.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6115

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PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-E-10062 - Engines: Preservation for Shipment and Storage of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-889 - Dissimilar metals.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13221E7330 - Generator Set, Diesel Engine, Trailer Mounted, 10 KW, 60 Hertz PU-753/M.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The PU-753/M shall be in accordance with TA13221E7330 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by

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the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.1.1.1 Vehicle marking. Registration numbers and other marking of the trailer shall be in accordance with MIL-STD-642 (see 6.7).

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions in 3.3.1, 3.3.2, and 3.3.3 of this specification.

3.3.1 Material deterioration and control. The item(s) shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage enviroments to which the item may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the generator set may be newly fabricated from recovered materials to the maximum extent practicable, provided the generator set produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the generator set.

3.4 Government-furnished property. The following property in the quantities indicated will be furnished by the Government (see 6.4):

Item No.	Description	Identification	Quantity for each PU-753/M
1	Chassis, Trailer, M116A2, 3/4-ton, 2-wheel.	NSN 2330-01-101-8434	1
2	Generator set, diesel-engine-driven, 10 kW, 60 Hertz DOD Model MEP-003A.	NSN 6115-00-465-1030	1

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3.5 Workmanship. Workmanship shall be of a quality to assure that the power units are free from defects resulting from defective material, incorrect manufacturing or assembly practices, incorrect treatment and painting, incomplete welds, rust, cracks, and other defects that could impair its operation or serviceability. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are specified on the applicable drawing.

3.6 Safety. All changes, deviations, and waivers to TA13221E7330 shall be in accordance with the following requirements. All rotating or reciprocating parts and other parts subject to high operational temperatures that are of such a nature or are so located as to be a hazard to operating personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or repeat any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Disassembly inspection. Failure of the examination by the first article model shall be cause for disassembly in the presence of a Government representative of the first article model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected. Reassembly with accepted parts and reexamination shall be the responsibility of the contractor. Government-furnished property shall be excluded from the requirements of this paragraph.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Acceptability criteria. Power units which conform to all requirements in sections 3 and 5 of this specification and pass all applicable examinations in section 4 of this specification will be considered acceptable by the Government.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article power unit shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each power unit shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The power units shall be examined for the following defects:

101. Materials not as specified.
102. Materials are resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the generator sets.
106. Treatment and painting not as specified.
107. Dimensions not as shown on drawings.
108. Components missing or not as specified.
109. Assembly not as shown on the drawings.
110. Workmanship not as specified.
111. Safety not as specified.

4.6 Inspection of packaging.

4.6.1 First article pack inspection. The first article pack shall be examined for defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection. Any defect(s) shall be corrected and the first article re-examined.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

112.	Generator sets not preserved as specified for levels A and B.
113.	Fire extinguisher not removed from bracket and preserved as specified for levels A and B.
114.	Power and ground cable not disconnected and secured as specified for levels A and B. Cable and terminals not matchmarked as specified.
115.	Bows not secured together as specified.
116.	Hammer, fuel can adapters, oil drain line and ground rods not secured as specified.
117.	Technical publications not preserved as specified for all levels.
118.	Tarpaulin not preserved as specified.
119.	Trailer chassis components not preserved as specified for levels A and B.
120.	Packing not as specified by the applicable figures for levels A and B.
121.	Tarpaulin roof supports not secured to unit as specified for levels A and B.
122.	Marking illegible, incomplete, incorrect, or missing.
123.	Depreservation guides not prepared and attached to the power unit as specified.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.5). The first article pack may be accomplished utilizing either the first article power unit or a production power unit. If the first article power unit is utilized, any preservation, and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article power unit and production power unit.

5.2 Preservation. Preservation shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Generator sets. The generator set shall be preserved in accordance with MIL-P-116, method IIa. The generator set shall be secured directly through the barrier material to the trailer chassis, using the required mounting bolts. Before positioning the barrier material over the generator, the generator shall be covered with a half slotted container constructed in accordance with PPP-B-636, without cover, type CF, class-domestic, variety SW, grade-200 minimum. The container shall be sealed in accordance with method I of the appendix thereto. All sharp edges of the container that can damage the barrier material shall be

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covered with non abrasive pressure sensitive tape. The top corners of the container shall be rounded to prevent them from puncturing the barrier material. Additionally, the bottom corners and the bottom portion of the forward and aft panels shall be trimmed to receive the wood wedges. The barrier shall be protected by a box-type cover consisting of ends, sides, and top constructed in accordance with PPP-B-601, for an overseas type, style I box. Sides and ends shall be 1/4-inch-thick plywood. The top shall be one piece 1/2- inch-thick plywood and shall completely cover all edges. The humidity indicator and inspection port shall be installed on an end or side in a location convenient for inspection of the humidity indicator. The cover shall be secured using wood wedges and lag screws. The wedges shall be cut to fit the angle of the ends of the generator skids and shall be of sufficient length and width to provide a snug fit to the inside of the cover. The angle surface of the wedge shall be cushioned with fiberboard where contact with the barrier is required. Lag screws shall be the minimum length required to secure the cover to prevent puncture of the barrier. Before inclosing the generator set in the Method IIa barrier, the following preservation shall be accomplished.

5.2.1.1.1 Engine. The engine and engine accessories shall be preserved in accordance with MIL-E-10062, level A, type II, method II. Drivebelts shall be relieved from tension and tagged and battery cables shall be disconnected and secured as specified therein.

5.2.1.1.2 Fire extinguisher. The fire extinguisher shall be removed from the bracket and preserved in a box conforming to PPP-B-636, type CF, class domestic, variety SW, grade 125, style optional and shall be secured to the generator set with tape conforming to PPP-T-97, type IV.

5.2.1.1.3 Cable. The ground cable shall be disconnected and the terminals on the generator set and cable ends shall be matchmarked with tags conforming to UU-T-81, type A. The cable shall be coiled, the coil secured. The coiled cable shall be secured to the generator set with tape conforming to PPP-T-97, type IV.

5.2.1.1.4 Hammer, fuel can adapters, oil drain line, and ground rods. The hammer, fuel can adapters, oil drain line, and ground rods shall be secured in the accessory box with tape conforming to PPP-T-97, type IV.

5.2.1.2 Trailer assembly.

5.2.1.2.1 Chassis. The trailer chassis components shall be preserved in accordance with MIL-STD-281, level A, as specified for mobile vehicles.

5.2.1.2.2 Bows. The bows shall be removed from the trailer pockets and secured together with strapping at six equally spaced positions. Wood blocking shall be used under the strapping to prevent damage to the bows. Strapping shall conform to QQ-S-781, class 1, type I or IV, size .020 x 3/4. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.1.2.2.1 Tarpaulin roof support. The tarpaulin roof support shall be removed and secured in any available space within the confines of the trailer.

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5.2.1.2.3 Tarpaulin. The tarpaulin shall be wrapped in film conforming to L-P-378, type I, class I, grade A, 4 to 8 mil and shall be preserved in a close-fitting box conforming to PPP-B-601, overseas type, style I. Strapping shall not be required.

5.2.1.3 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 and shall be secured to the generator set outside the method II barrier.

5.2.2 Level B. The generator set shall be preserved as specified for level A except for the following.

5.2.2.1 Box-type cover. The box-type cover specified in 5.2.1.1 shall be domestic type.

5.2.2.2 Engine. The engine and engine accessories shall be preserved in accordance with MIL-E-10062, level A, type II, method I.

5.2.2.3 Strapping. The strapping specified in 5.2.1.2.2 shall be finish A.

5.2.2.4 Tarpaulin. The box specified in 5.2.1.2.3 shall be domestic type.

5.2.2.5 Trailer chassis. The trailer chassis components shall be preserved in accordance with MIL-STD-281, level B, as specified for mobile units.

5.2.3 Level C. Power units including all components and surfaces requiring it, shall be preserved to provide protection against deterioration and damage during shipment and handling from the contractor to the initial destination. Cables, fuel, and oil drain lines shall not be disconnected. Technical publications shall be placed in a waterproof bag and secured to the generator set.

5.3 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.3.1 Level A. The bows shall be placed and secured on the power unit with the solid area of the bows facing the front panel of the trailer. The tarpaulin box shall be secured to the forward end of the generator box cover with two horizontal metal straps, each encompassing both boxes. Blocking shall be provided to support and aid in securing the bows in position and to prevent the tarpaulin box from raising up. Strapping for securing the bows and tarpaulin shall conform to QQ-S-781, class I, type I or IV, size 0202. x 3/4. Unless otherwise specified (see 6.2), strapping shall be finish B.

5.3.2 Level B. Packing shall be as specified for level A, except that strapping specified in 5.3.1 securing the bows, tarpaulin, and parts shall be finish A.

5.3.3 Level C. Power units shall be completely mobile. Any components not placed in secure brackets shall be removed from them and placed in an out of the way area and secured to prevent movement. The tarpaulin cover shall be in place upon its bows.

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5.4 Marking. Marking for all levels shall be in accordance with MIL-STD-129.

5.5 Depreservation guides. Two copies of depreservation guides shall be prepared and placed in waterproof envelopes marked "Depreservation Guide". One copy shall be secured on the generator set and a second copy shall be secured in a conspicuous location on the power unit. Unless otherwise specified (see 6.2), DA Form 2258 "Depreservation Guide for Vehicles and Equipment" shall be used (see 6.6).

5.6 Preparation of Government-furnished property. Government-furnished property shall be preserved, packaged, and packed as specified herein, regardless of the preservation, packaging, and packing provided at the time of receipt by the contractor.

6. NOTES

6.1 Intended use. The power unit is intended for use by the armed services to supply electrical power for military operations.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. Time frame required for submission of the first article model (see 3.2).
d. Time frame required for submission of the first article pack (see 5.1).
e. Degree of preservation and of packing required (see 5.2 and 5.3).
f. When other than DA Form 2258 is to be used (see 5.5).

6.3 First article. When a first article inspection is required, the items should be a first article model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first test results and disposition of the document's first articles.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.4.

6.5 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the power units in accordance with this specification.

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6.6 Depreservation guide. The contracting officer should arrange to furnish the necessary copies of DA Form 2258, when requested by the contractor (see 5.5).

6.7 Registration numbers. The contracting officer should arrange to furnish a list of registration numbers to be applied to the trailer (see 3.1.1.1).

Custodian:
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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-52903A(ME)	Generator Set, Diesel Engine, Trailer Mounted 10 KW, 60 Hertz,

3a. NAME OF SUBMITTING ORGANIZATION PU-753/M	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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